A Framework for Categorizing Complaint Text via Large Language Model

Liu Yuan[1,2,a], Xiaoye Ouyang[1,b], Ran Bai[1,2,c], Xiaoling Zhu[1,2,d], Yanguo Zhou[1,2,e], Yi Zhang[1,2,f], Chenyu Liu[1,2,g*]
[1]China Academy of Electronic and Information Technology, Beijing, China
[2]CETC Academy of Electronics and Information Technology Group Co., Ltd, Beijing, China
[a]yuanliu@cete.com.cn; [b]ouyangxiaoye@cete.com.cn; [c]ran.bai@outlook.com; [d]zhuxiaoling1@cetc.com.cn;
[e]zhouyanguo@cetc.com.cn; [f]iharrycuplcetc@163.com; [g*]Corresponding author: liuchenyu@cete.com.cn

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***Abstract*—With the development of the Internet, businesses are confronted with a burgeoning volume of cross-industry complaint information, posing challenges for traditional manual classification methods. In this paper, a Complaint Text Classification Framework (CCF) is proposed, leveraging state-of-the-art Large Language Models to categorize complaint information efficiently. CCF adopts a multi-step approach, including real-time monitoring, data preprocessing, feature extraction, and classification prediction, with the aim of enhancing the accuracy and efficiency of complaint categorization for organizations grappling with vast datasets. Extensive experimental evaluations demonstrate that bolstered by the utilization of the Yi-34B large model, CCF excels in accurately categorizing complaints, underscoring its potential to streamline complaint management processes within today's data-driven environment. This framework also serves as a benchmark for future research endeavors.**

Keywords-complaint information, text classification, prompt, bert, large language model

I. INTRODUCTION

As the Internet expands, businesses increasingly rely on user feedback to enhance products and services. Complaints, a vital component of this feedback, convey users' dissatisfaction[1]. However, as businesses grow, the volume and diversity of complaints surge, presenting challenges for traditional manual classification methods, which suffer from low efficiency, subjectivity, and inconsistent standards[2].

Previous research has proposed utilizing machine learning methods for classifying various complaint information. However, these approaches encounter practical limitations[3], [4]. Firstly, manually annotated datasets entail significant time and labor costs, especially when dealing with diverse and time-sensitive complaint information. Secondly, machine learning models typically require training on large-scale datasets, placing high demands on computational resources[5]. Lastly, these methods exhibit limited capabilities in processing long texts, semantic understanding, and contextual comprehension, particularly when considering multiple domains and dimensions in complaint information.

To address these limitations, this paper introduces the Complaint Text Classification Framework (CCF) based on large-scale language models (LLMs). The framework encompasses real-time monitoring, data preprocessing, feature extraction, and classification prediction. Through these steps, CCF achieves intelligent and efficient complaint classification, leveraging the advanced semantic understanding capabilities of large-scale language models and the pre-trained language model BERT. Compared to previous methods, CCF's primary contributions are highlighted as follows:

(1) Automating feature selection and model tuning to enhance adaptability to evolving complaint data.

(2) Overcoming constraints in handling the diversity and complexity of large complaint datasets.

(3) Improving classification efficiency and consistency, is crucial for managing the escalating volume and variety of complaints amidst expanding business domains.

II. RELATED WORK

A. Large Language Models

Large Language Models (LLM) refer to a natural language processing model with a massive parameter scale and powerful learning capabilities[6]. Its development originated from early word embedding models, such as Word2Vec[7] and GloVe[8], which primarily focused on mapping vocabulary to vector space but had limited expressive capabilities. With the rise of deep learning, large-scale pre-trained language models have become a trend, with models like GPT and BERT leading the way. These models are pre-trained on large-scale corpora, acquiring rich language knowledge and demonstrating outstanding performance across various NLP tasks[9].

Today, existing LLMs have evolved into Transformer language models with billions (or even more) of parameters, trained on massive text data, including GPT-3, PaLM, Galactica, and LLaMA. Recently, prominent attention has been given to large language models represented by ChatGPT. To address the time-consuming and labor-intensive nature of manual data annotation, researchers have explored methods based on large language models for few-shot or zero-shot annotation[10]. Wei et al.[11] introduced ChatIE, a multi-turn question-answering framework based on ChatGPT, aimed at zero-shot information extraction in general domain text, achieving significant results. Gilardi et al.[12] found that ChatGPT exhibits better annotation performance in tasks like relevance, stance, topic, and frame detection, with annotation costs per instance approximately 20 times cheaper than the MTurk platform.

Currently, LLMs have demonstrated excellent efficiency in text classification and robust semantic understanding. In this paper, we propose using an LLM (Yi-34B) for feature extraction

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in complaint text, simplifying long text sentences for subsequent classification tasks.

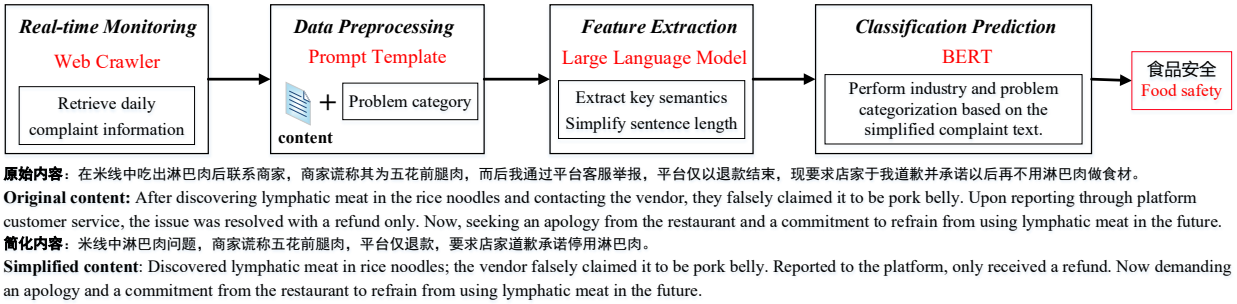
B. Text Classification

With the widespread application of deep learning in fields like image processing and machine translation, researchers have begun to leverage the high-feature extraction capabilities of deep learning for text classification[13]. For example, Marivate et al. [14] proposed an R-Transformer_BiLSTM model based on label embedding and attention mechanism for multi-label text classification. However, when dealing with long sequences, an increased gap between sequence data points leads to a weakening of the memorization ability in Long Short-Term Memory (LSTM) networks[15]. Additionally, models proposed by previous researchers often neglect the structured features inherent in long texts, resulting in low accuracy in long text classification and an increase in machine hardware overhead.

In response to the challenges, the BERT model, with features such as bidirectional contextual understanding and handling complex vocabulary through subword embeddings, has become a versatile and high-performance text classification tool[16]. Additionally, by incorporating attention mechanisms at both the word and sentence levels, it is possible to effectively enhance the accuracy of text classification. However, complaint texts are typically long, presenting challenges for BERT models[17]. To overcome these limitations, the text can be segmented and input into the BERT model, with the output word vector results serving as the data source for subsequent Transformer models. Furthermore, the information at both the word and sentence levels often lacks the complete semantic information of an entire sentence. Therefore, our proposed framework utilizes a large language model to extract key information from long complaint texts, preserving crucial semantic information while shortening sentence length, and applies this extracted data to BERT models for classification predictions.

III. METHODOLOGY

The CCF framework is depicted in Fig. 1.



原始内容：在米线中吃出淋巴肉后联系商家，商家谎称其为五花前腿肉，而后我通过平台客服举报，平台仅以退款结束，现要求店家于我道歉并承诺以后再不用淋巴肉做食材。
Original content: After discovering lymphatic meat in the rice noodles and contacting the vendor, they falsely claimed it to be pork belly. Upon reporting through platform customer service, the issue was resolved with a refund only. Now, seeking an apology from the restaurant and a commitment to refrain from using lymphatic meat in the future.
简化内容：米线中淋巴肉问题，商家谎称五花前腿肉，平台仅退款，要求店家道歉承诺停用淋巴肉。
Simplified content: Discovered lymphatic meat in rice noodles; the vendor falsely claimed it to be pork belly. Reported to the platform, only received a refund. Now demanding an apology and a commitment from the restaurant to refrain from using lymphatic meat in the future.

Fig. 1. Illustration of the proposed CCF framework for complaint classification.

A. Real-time Monitoring

To continuously track the influx of complaint information and ensure timely responses to new data, the monitoring module regularly accesses the platform. Real-time monitoring module utilizes *Selenium* and *Beautiful Soup* to crawl real-time complaint data from the "Black Cat Complaints" platform in China. Supported by *Selenium*, the web crawler simulates user browsing actions to load pages and extract HTML source code. *Beautiful Soup* parses the HTML to retrieve core data, such as Chinese complaint content and dates, providing rich data sources for subsequent text processing and platform maintenance.

B. Data Preprocessing

This module preprocesses complaint text by cleaning and standardizing it and removing special characters, punctuation, and HTML tags for consistency. It also involves tokenization, lemmatization, and stop-word removal to maintain text consistency. Additionally, the quality of prompt templates significantly affects model outputs, particularly in handling semantic ambiguity. Hence, we designed prompt structures to align with those predefined as follow:

"你作为智能语言总结工具, 帮助我提炼下述内容的主要思路和脉络, 并围绕主题生成一段简明扼要的总结。

【投诉文本内容】

As an intelligent language summarization tool, helps me distill the main ideas and structure from the following content, and generate a concise summary around the theme.

[Complaint text content] "

C. Feature Extraction

Previous studies often neglect the structural features in lengthy texts, leading to lower classification accuracy and increased hardware costs. Word and sentence-level information often can't fully grasp the text's complete meaning. Our framework uses large language models to extract crucial information from long complaint texts, preserving essential meaning while shortening sentences.

The choice of a LLM significantly impacts the feature extraction module's performance. While commercial models are powerful, they're costly and have limited daily token analysis. However, open-source models are closing the performance gap. Ultimately, we chose the Yi-34B model for simplifying complaint information, known for its outstanding performance in the hugging-face Chinese ranking.

During this step, we input preprocessed complaint texts into the language model, directly leveraging its robust representation capability to extract feature vectors. This process captures key semantic features without additional engineering. These vectors

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can be directly used for downstream classification tasks, to enhance prediction accuracy and efficiency.

D. Classification Prediction

Given BERT's superior capacity to understand contextual and semantic nuances, we employ it to classify simplified complaint texts. Compared to conventional methods like SVM and Naive Bayes, BERT excels in capturing semantic intricacies and contextual connections, overcoming challenges with dimensionality and lengthy texts. This enhances classification accuracy and efficiency.

For a dataset D containing n complaint information samples, the expression is (1):

$$D = \{C_i \mid i = 1, 2, 3, ..., m\}, \sum_{i=1}^{m} \mid C_i \mid = n \tag{1}$$

Where represents the category of textual data, and denotes the total number of texts in the category C_i. The main steps of the classification process are as follows:

(1) In the tokenization process, handling stop words by deleting certain meaningless words or markers to simplify complaint text.

(2) For short text data $d = \{w_1, w_2, ..., w_N\}$, which represents the text token, according to the BERT model, the vector representation of the text $d([CLS])$ and its corresponding token w_i is obtained as (2):

$$\begin{cases} d' = \{b_1, b_2, ..., b_M\}, \\ \overrightarrow{w_i} = b_{i1}, b_{i2}, ..., b_{iM} (i = 1, 2, 3, ..., N), \end{cases} \tag{2}$$

In this expression, M represents the length of the text token vector, which is the *hidden_states* status output from the last layer of the BERT model, i.e., *hidden_size*.

(3) Obtain the mean vector representation of text tokens. Based on the output text, the expression is given by:

$$\begin{cases} d'_{mean} = \{b'_1, b'_2, ..., b'_N\}, \\ b'_j = \dfrac{1}{N} \sum_{i=1}^{N} b_{ij} (j = 1, 2, ..., 768). \end{cases} \tag{3}$$

(4) Through the *Fully Connected layer* (FC) and *Softmax* processing, the final classification result is obtained, and the output obtained through the fully connected layer is:

$$y = W^T d' + bias \tag{4}$$

Where W is a $768 \times m$ -dimensional weight matrix, and $bias$ is the bias term, given by:

$$y'_{\mathrm{k}} = Soft\max(y_k) = \frac{e^{yk}}{\sum_{j=1}^{m} e^{yj}}, \sum_{k=1}^{m} y'_{\mathrm{k}} = 1. \tag{5}$$

(5) Based on the value of the output y'_k , obtain the classification label for the complaint text d , and apply the same process for problem category classification.

IV. Experiments

A. Dataset

To assess the efficacy of the framework, we curated a dataset consisting of complaint records from February 1, 2024, totaling 14,914 entries. The primary complaint categories encompass Cosmetics, Drugs and Medical Devices, False Advertising, Quality Issues, Food Safety, and Delivery Rider. The detailed classification results are presented in Table Ⅰ. Importantly, these data underwent meticulous manual classification by five human annotators, ensuring a high degree of accuracy in labeling. These annotated data serve as a benchmark for subsequent model validation.

TABLE I. Details of the experimental dataset

Categories Labels	COMPLAINT PROBLEM CATEGORIES	
	Categories	*Quantity*
0	其他 (Other)	7,617
1	化妆品 (Cosmetics)	280
2	药械(Drugs and Medical Devices)	232
3	虚假宣传 (False Advertising)	1,084
4	质量问题 (Quality Issues)	2,858
5	食品安全 (Food Safety)	471
6	骑手 (Delivery Rider)	2,086

B. Evaluation Metric

The accuracy is used to evaluate the performance of the classification model. A higher accuracy score indicates that the model more accurately assigns complaint information to their respective categories. The formula for calculating accuracy is as follows:

$$\textbf{\textit{Accuracy}} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Instances}} \tag{6}$$

C. Baseline Models

To validate the effectiveness of the CCF framework, we conducted comparisons with traditional text classification methods and those based on pre-trained language models. This evaluation aims to assess the performance of the CCF in handling complaint information classification tasks:

1) CNN[18]: This is a baseline approach in text classification, using a one-layer CNN architecture.

2) BiLSTM[19]: This method employs Bidirectional Long Short-Term Memory (BiLSTM) and an attention mechanism to encode a sentence for classification.

3) BERT[20]: This method initially acquires the hidden embedding of the [CLS] token through BERT and subsequently feeds this embedding into the classification layer for prediction.

4) GPT-3.5-Turbo: It's an advanced language model developed by OpenAI, capable of diverse natural language processing tasks with improved performance.

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D. Experiment Results

TABLE II. THE ACCURACY OF MODEL CLASSIFICATION

Categories Labels	The accuracy of model classification				
	CNN	BiLSTM	BERT	GPT-3.5-Turbo	CCF
1	0.6575	0.6638	0.7114	0.7268	0.9845
2	0.6177	0.6416	0.6436	0.6526	0.8065
3	0.6620	0.6867	0.7760	0.7950	1.0000
4	0.7098	0.7177	0.7907	0.9800	1.0000
5	0.6585	0.6720	0.7901	0.9050	0.9900
6	0.6406	0.6689	0.7786	0.8797	0.9699

[a.] Bold values indicate the optimal results.

The experimental results highlight the strong performance of various classification methods on diverse complaint datasets, as shown in Table Ⅱ. BERT outperforms traditional approaches like CNN and BiLSTM, thanks to its superior feature representation capabilities. GPT-3.5-Turbo also shows impressive comprehension abilities and knowledge across classification tasks. Our CCF framework consistently achieves optimal results in each category. However, in datasets with multiple categories, the predictive performance for pharmaceutical complaints is relatively lower, likely due to their complexity and overlap with the cosmetics category.

Further analysis reveals that the CCF framework excels by combining the feature extraction of large language models (LLM) with the classification efficiency of BERT. Unlike direct classification methods used by BERT and GPT-3.5-Turbo, the integrated approach in the CCF framework enhances precision in complaint content classification.

E. The Impact of LLMs

We explore the impact of different Large Language Models (LLMs) on overall classification performance. We replace the LLM in the third step of the framework with various models, each sourced from https://huggingface.co, and the experimental results are shown in Fig 2. Detailed information for each LLM is provided below:

1) ERNIE-Bot-turbo: A large-scale Chinese language model developed by Baidu, excelling in conversational question-answering and content generation.

2) AquilaChat2-7B: An open-source chat model with 7 billion parameters, designed to enhance human interactions.

3) Spark Desk by iFLYTEK: An intelligent large-scale model software, integrating natural language understanding and logical reasoning for interactive dialogue.

4) Yi-34B: A large pre-trained language model with 340 billion parameters, known for its exceptional performance in Hugging Face Chinese rankings and open-source capabilities.

The experiments underscore the crucial role of different LLMs in feature extraction from complaint texts, directly impacting BERT model classification accuracy. ERNIE-Bot-turbo and Spark Desk significantly enhance semantic understanding and generation capabilities, yielding practical results. However, AquilaChat2-7B, while employing a similar approach, exhibits lower generation performance, affecting classification outcomes. Meanwhile, Yi-34B focuses on enhancing language generation quality and diversity, leveraging attention mechanisms and generative adversarial networks to achieve outstanding performance in classifying complaint texts.

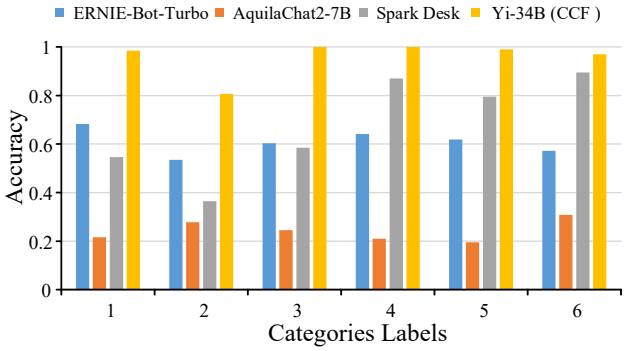


Fig. 2. Impact of different large language models on overall classification performance.

V. CONCLUSION

In this paper, we present the Complaint Classification Framework (CCF), which utilizes Large Language Models for effective categorization of cross-industry complaint data. CCF, powered by the Yi-34B model, involves real-time monitoring, data preprocessing, feature extraction, and classification prediction. This framework enhances complaint processing efficiency and accuracy. Experimental results demonstrate CCF's superiority in handling diverse complaint datasets compared to previous methods. In future work, we will concentrate on advanced feature extraction techniques to bolster CCF's capability in identifying intricate patterns within complaint data. Additionally, we also intend to broaden CCF's reach to encompass multilingual scenarios, aiming to enhance its applicability across diverse industries and regions and ultimately improve complaint management systems.

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